



Image may differ from product. See technical specification for details.

2204 E-2RS1TN9

Self-aligning ball bearing with seals on both sides

Self-aligning ball bearings, with seals on both sides, have two rows of balls, a common sphered raceway in the outer ring and two deep uninterrupted raceway grooves in the inner ring. They are insensitive to angular misalignment of the shaft relative to the housing. The integral sealing can significantly prolong bearing service life because it keeps lubricant in the bearings and contaminants out.

- Accommodate static and dynamic misalignment
- Excellent high-speed performance
- Excellent light load performance
- Low friction
- Integral sealing results in reduced maintenance requirements and prolonged bearing service life

Overview

Dimensions

Bore diameter	20 mm
Outside diameter	47 mm
Width	18 mm

Performance

Basic dynamic load rating	12.7 kN
Basic static load rating	3.4 kN
Reference speed	28 000 r/min
Limiting speed	10 000 r/min

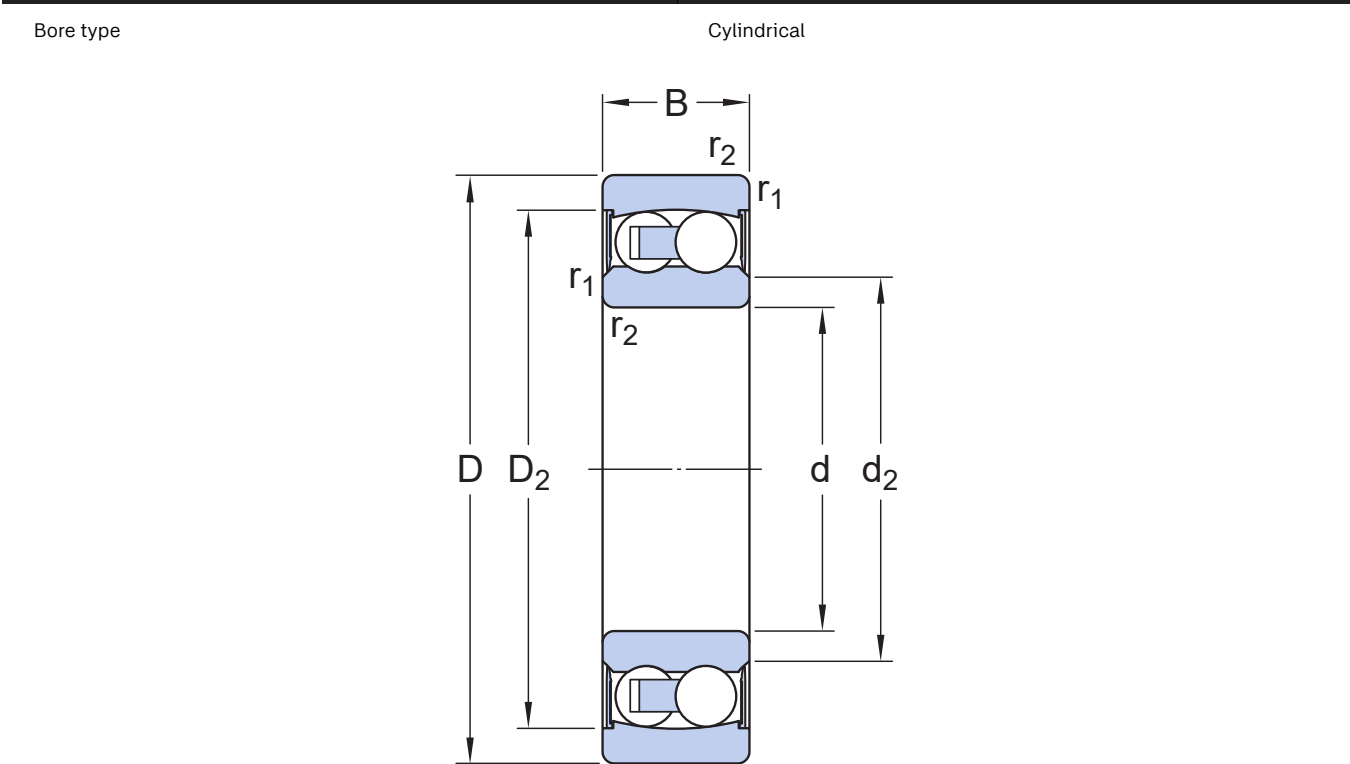
Properties

Retaining feature, inner ring	None
Locating feature, bearing outer ring	Without
Number of rows	2
Bore type	Cylindrical
Cage	Non-metallic
Radial internal clearance	CN
Tolerance class	Normal
Material, bearing	Bearing steel
Coating	Without
Sealing	Seal on both sides
Sealing type	Contact
Lubricant	Grease
Relubrication feature	Without
Indicative carbon footprint for new product	0.5 kg CO ₂ e

Logistics

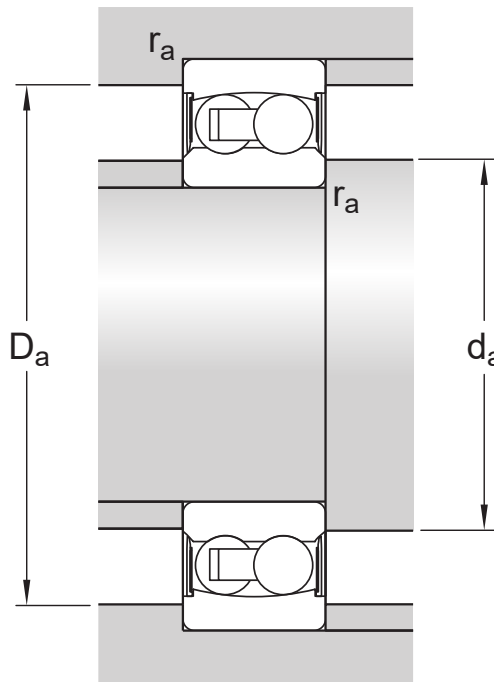
Product net weight	0.139 kg
eClass code	23-05-08-06
UNSPSC code	31171532

Technical specification



Dimensions

d	20 mm	Bore diameter
D	47 mm	Outside diameter
B	18 mm	Width
d ₂	≈ 25.9 mm	Recess diameter inner ring
D ₂	≈ 41.23 mm	Recess diameter outer ring
r _{1,2}	min. 1 mm	Chamfer dimension



Abutment dimensions

d _a	min. 25 mm	Abutment diameter shaft
d _a	max. 25.5 mm	Abutment diameter shaft
D _a	max. 41.4 mm	Abutment diameter housing
r _a	max. 1 mm	Fillet radius

Calculation data

Basic dynamic load rating	C	12.7 kN
Basic static load rating	C ₀	3.4 kN
Fatigue load limit	P _u	0.176 kN
Reference speed		28 000 r/min
Limiting speed		10 000 r/min
Permissible angular misalignment	α	1.5 °
Calculation factor	k _r	0.05
Limiting value	e	0.3
Calculation factor	Y ₀	2.2
Calculation factor	Y ₁	2.1
Calculation factor	Y ₂	3.3

Tolerances and clearances

GENERAL BEARING SPECIFICATIONS

- [Tolerances: Normal, JS7](#)
- [Radial internal clearance: table](#)

BEARING INTERFACES

- [Seat tolerances for standard conditions](#)
- [Tolerances and resultant fits](#)



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